

External Controls — Floor Shift

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Inspection and Verification

1. Verify the customer concern by operating the transmission external control.
2. Visually inspect for obvious signs of mechanical and electrical damage; refer to the following chart:

Visual Inspection Chart

Mechanical	Electrical
• Damaged Brake Shift Interlock Actuator (BSIA) • Damaged Tow/Haul switch • Damaged SelectShift™ switch	• Failed fuse(s): ▪ Body Control Module (BCM) 18 (10A) ▪ BCM 42 (5A) • PCM • Loose or damaged connections • Damaged wiring harness • Corroded or damaged terminals

3. If the fault is not visually evident, determine the symptom. GO to [Symptom Chart — External Controls, Floor Shift](#) or GO to [Symptom Chart — NVH Floor Shift](#).

DTC Chart

DTC Chart

DTC	Component	Description	Condition	Symptom	Action
B1319:11	Brake Shift Interlock Actuator (BSIA) , BCM	Shift Interlock (BSI) Output Circuit: Circuit short to ground	BSIA system malfunction	BSIA inoperative.	GO to Pinpoint Test E.
B1319:15	Brake Shift Interlock Actuator (BSIA) , BCM	Shift Interlock (BSI) Output Circuit: Circuit short to voltage or open	BSIA system malfunction	BSIA inoperative.	GO to Pinpoint Test E.
P0815	Upshift Switch Circuit	Upshift switch concerns, open or shorted circuit	The PCM signal input voltage status is out of range	Upshift switch inoperative.	GO to Pinpoint Test H.
P0816	Downshift Switch Circuit	Downshift switch concerns, open or shorted circuit	The PCM signal input voltage status is out of range	Downshift switch inoperative.	GO to Pinpoint Test H.
P1780	Tow/Haul Switch	Transmission Control Switch (TCS) (Overdrive (O/D) cancel) circuit out of self-test range	The PCM did not receive a Tow/Haul switch input during the Key ON Engine Running (KOER) self-test	Tow/Haul is inoperative or does not operate correctly.	GO to Pinpoint Test G.

Symptom Chart — External Controls, Floor Shift

Symptom Chart — External Controls, Floor Shift		
Condition	Possible Causes	Action
The Brake Shift Interlock Actuator (BSIA) system does not release/lock correctly	- Circuitry - Fuse(s): - Body Control Module (BCM) 18 (10A) - BSIA - Selector lever cable	GO to Pinpoint Test E. - VERIFY selector lever cable adjustment. REFER to Selector Lever Cable Adjustment — Floor Shift in this section.
The shift control is out of correct gear relationship	- Selector lever cable and bracket - Selector lever linkage	GO to Pinpoint Test F.
Transmission Range (TR) indicator (PRNDL) does not correspond to the selected gear	Selector lever cable bracket	TIGHTEN bolts holding selector lever cable bracket. REFER to Selector Lever Cable Adjustment — Floor Shift in this section.

	• Selector lever cable loose at the selector lever • Selector lever cable	• SECURE selector lever cable. REFER to Selector Lever Cable Adjustment — Floor Shift in this section. • VERIFY selector lever cable adjustment. REFER to Selector Lever Cable Adjustment — Floor Shift in this section.
• TR indicator (PRNDL) does not illuminate	• Instrument Panel Cluster (IPC) malfunction • Circuitry • TR sensor	• REFER to Section 413-00. • CHECK for TR sensor DTCs. REFER to Section 307-01.
• The ignition key cannot be returned to the OFF position or the ignition key can be turned to the OFF position when the selector lever is not in PARK	• Wiring, terminals or connectors • Ignition lock cylinder • Key removal inhibit solenoid (part of the ignition switch) • Park detect switch (part of the selector lever) • Selector lever cable • Mechanical damage to the selector lever, selector lever cable or selector lever knob	• REFER to Section 211-05.
• The PRNDL indicator does not illuminate P	• Wiring, terminals or connectors • PCM concern • IPC • Park detect switch (part of the selector lever)	• REFER to Section 413-01.
• The Tow/Haul switch does not operate correctly	• Tow/Haul switch • Tow/Haul switch not cycled during self-test • Circuitry	• GO to Pinpoint Test G.

• The Tow/Haul lamp is not operating correctly	• IPC • PCM • Circuitry	• REFER to Section 413-01.
• The SelectShift™ switch is not operating correctly	• SelectShift™ switch • PCM • Circuitry	• GO to Pinpoint Test H.
• Water enters inside the vehicle	• Cable assembly grommet • Torn selector lever cable assembly grommet	• SECURE grommet to floor panel. REFER to Selector Lever Cable — Floor Shift in this section. • INSTALL a new selector lever cable. REFER to Selector Lever Cable — Floor Shift in this section.
• Excessive shift effort	• Selector lever cable • Cable bracket	• INSTALL a new selector lever cable. REFER to Selector Lever Cable — Floor Shift in this section. • ADJUST cable. TIGHTEN cable bracket screws. REFER to Selector Lever Cable — Floor Shift in this section.
• Selector lever will not shift from PARK	• Selector lever cable • Selector lever knob • Broken selector lever cable • Brake Shift Interlock Actuator (BSIA) system failure	• VERIFY selector lever cable adjustment. REFER to Selector Lever Cable Adjustment — Floor Shift in this section. • INSTALL a new selector lever knob. REFER to Selector Lever Knob in this section. • INSTALL a new selector lever cable. REFER to Selector Lever Cable — Floor Shift in this section. • GO to Pinpoint Test E.
• Vehicle will not crank/start in park but will crank/start in neutral	• Selector lever cable	• VERIFY selector lever cable adjustment. REFER to Selector Lever Cable

	• TR sensor	Adjustment — Floor Shift in this section. • CHECK for TR sensor DTCs. REFER to Section 307-01.
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Symptom Chart — NVH Floor Shift

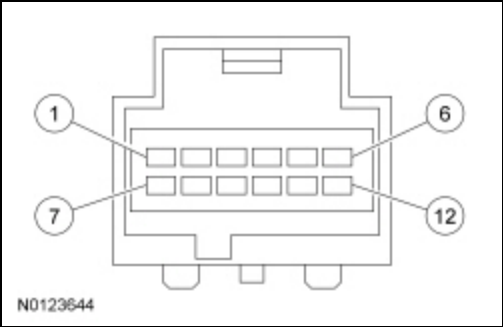
Symptom Chart — NVH Floor Shift		
Condition	Possible Causes	Action
NOTE: NVH symptoms should be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to Section 100-04. Since it is possible any one of multiple systems may be the cause of a symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to Section 100-04 for the next likely system and continue diagnosis.		
• Vibration — a high frequency (20-80 Hz) that is felt through the seat or selector lever. Changes with engine speed	• Selector lever cable incorrectly routed, grounded out or loose	• CHECK the selector lever cable. REPAIR as necessary. REFER to Selector Lever Cable — Floor Shift in this section.
• Rattle, buzz or other noise	• Selector lever loose • Selector lever knob • Selector lever • Cable assembly grommet	• TIGHTEN the selector lever bolts. REFER to Selector Lever — Floor Shift in this section. • INSTALL a new selector lever knob. REFER to Selector Lever Knob in this section. • INSTALL a new selector lever. REFER to Selector Lever — Floor Shift in this section. • SECURE the grommet to the floor panel. REFER to Selector Lever Cable — Floor Shift in this section.

Pinpoint Tests

Refer to Wiring Diagrams Cell [30](#) , Transmission Controls for schematic and connector information.

Refer to Wiring Diagrams Cell [37](#) , Shift Interlock for schematic and connector information.

Selector Lever Connector — C3245



Pin Number	Circuit Function
1	BSIA Power
2	Key Lock Solenoid (inhibit removal)
3	Transmission Park Detect
4	Illumination
5	Ground
6	Tow/Haul Power
7	Tow/Haul Signal
8	Not Used
9	SelectShift™ Upshift
10	SelectShift™ Downshift
11	SelectShift™ Ground
12	Not Used

PINPOINT TEST E : THE BSIA SYSTEM DOES NOT RELEASE/LOCK CORRECTLY

NOTE: *Inspect the selector lever cable before proceeding with this pinpoint test.*

E1 CHECK THE STOPLAMPS

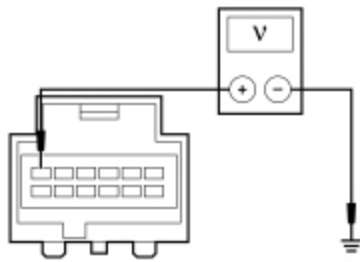
- Apply the brake pedal and observe the stoplamps.

Do the stoplamps operate correctly?

Yes	GO to E2 .
No	REFER to Section 417-01.

E2 CHECK THE BSIA POWER CIRCUIT FOR VOLTAGE

- Disconnect: Floor Shifter [C3245](#).
- Ignition ON.
- Measure the voltage between the BSIA [C3245](#) Pin 1, circuit CET53 (BU/OG), harness side and ground.



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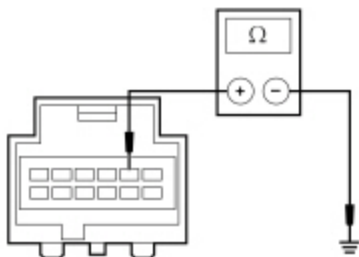
- Apply the brake pedal.
- With the brake pedal applied, measure the voltage between the BSIA [C3245](#) Pin 1, circuit CET53 (BU/OG), harness side and ground.

Is the voltage greater than 10 volts with the brake pedal applied and 0 volts with the brake pedal not applied?

Yes	GO to E3 .
No	VERIFY the BCM fuse 18 (10A) is OK, If OK REPAIR circuit CET53 (BU/OG) for an open. CLEAR the DTC. TEST the system for normal operation. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. If voltage was present with the brake pedal not applied, GO to E4 .

E3 CHECK THE BSIA GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the resistance between the selector lever [C3245](#) Pin 5, circuit GD138 (BK/WH), harness side and ground.



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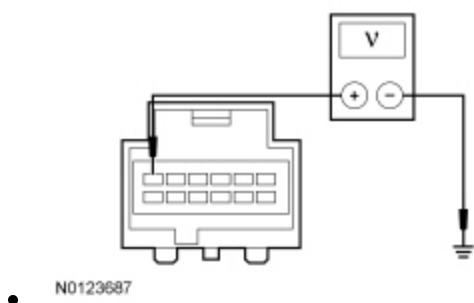
Is the resistance less than 5 ohms?

Yes	INSTALL a new selector lever. REFER to Selector Lever — Floor Shift in this section. CLEAR the DTC. TEST the system for normal operation.
No	REPAIR circuit GD138 (BK/WH) for an open. CLEAR the DTC. TEST the system for normal operation.

E4 CHECK THE BSIA OUTPUT CIRCUIT FOR A SHORT TO POWER

- Disconnect: BCM [C2280B](#) .
- Inspect [C3245](#) and [C2280B](#) for damaged, pushed-out terminals or loose wires.

- Ignition ON.
- Measure the voltage between the selector lever [C3245](#) Pin 1, circuit CET53 (BU/OG), harness side and ground.



Is any voltage present?

Yes	REPAIR circuit CET53 (BU/OG) for a short to power. CLEAR the DTC. TEST the system for normal operation.
No	GO to E5 .

E5 CHECK FOR CORRECT BCM OPERATION

- Disconnect all **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . CLEAR the DTC. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

PINPOINT TEST F : THE SHIFT CONTROL IS OUT OF CORRECT GEAR RELATIONSHIP

F1 CHECK THE SHIFT CONTROL LINKAGE

- Ignition ON.
- Apply the brake pedal.
- Gain access to the selector lever linkage.
- Position the selector lever in all ranges.
- Observe all selector lever linkage during operation.

Is the selector lever linkage damaged?

Yes	INSTALL a new selector lever. REFER to Selector Lever — Floor Shift in this section. TEST the system for normal operation.
No	GO to F2 .

F2 CHECK THE SELECTOR LEVER CABLE

- Check the selector lever cable and bracket installation and tightness.

Is the selector lever cable correctly installed and adjusted?

Yes	VERIFY the selector lever cable adjustment. ADJUST the selector lever cable if necessary. REFER to Selector Lever Cable Adjustment — Floor Shift in this section. GO to F3 .
No	REPAIR as necessary. TEST the system for normal operation.

F3 CHECK THE LINKAGE/CABLE FOR CORRECT GEAR RELATIONSHIP

- Apply the brake pedal.
- Position the selector lever in all ranges.

Does the indicator match the gear selection?

Yes	System OK. TEST the system for normal operation.
No	REFER to Selector Lever Cable — Floor Shift in this section. TEST the system for normal operation.

PINPOINT TEST G : THE TOW/HAUL SWITCH DOES NOT OPERATE CORRECTLY

NOTE: Vehicles equipped with a 3.7L engine use PCM [C175T](#).

NOTE: Vehicles equipped with a 5.0L or 6.2L engine use PCM [C1381T](#).

NOTE: Vehicles equipped with a 3.5L engine use PCM [C1551B](#).

G1 CHECK DTCS

- Ignition ON.
- Connect the scan tool.
- Start the engine and RUN the Key ON Engine Running (KOER) self-test.

Is DTC P1780 set?

Yes	GO to G2 .
No	No trouble found with the Tow/Haul system. REFER to Section 413-01 if Tow/Haul lamp does not illuminate.

G2 CHECK THE BCM FUSE 42 (5A)

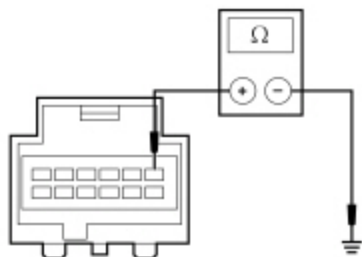
- Ignition OFF.
- Remove Body Control Module (BCM) fuse 42 (5A).
- Check fuse: BCM 42 (5A) .

Is the resistance less than 5 ohms?

Yes	GO to G4 .
No	GO to G3 .

G3 CHECK THE TOW/HAUL POWER CIRCUIT FOR A SHORT TO GROUND

- Disconnect: Selector Lever [C3245](#).
- Disconnect: BCM [C2280B](#) .
- Inspect [C3245](#) and [C2280B](#) for damaged, pushed-out terminals or loose wires.
- Measure the resistance between the selector lever [C3245](#) Pin 6, circuit CBP42 (GN), harness side and ground.



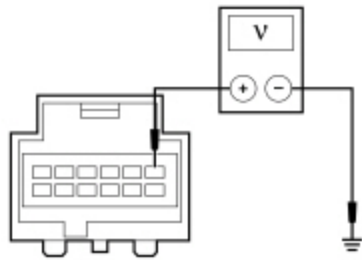
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Is the resistance greater than 10,000 ohms?

Yes	GO to G4 .
No	REPAIR circuit CBP42 (GN) for a short to ground. CLEAR the DTC. RERUN the KOER self-test and test the system for normal operation.

G4 CHECK THE TOW/HAUL POWER CIRCUIT FOR VOLTAGE

- Disconnect: Selector Lever [C3245](#).
- Inspect [C3245](#) for damaged, pushed-out terminals or loose wires.
- Ignition ON.
- Measure the voltage between the selector lever [C3245](#) Pin 6, circuit CBP42 (GN), harness side and ground.



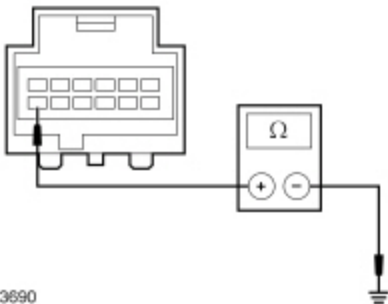
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Is the voltage greater than 10 volts?

Yes	GO to G5 .
No	REPAIR circuit CBP42 (GN) for an open. CLEAR the DTC. RERUN the KOER self-test and test the system for normal operation.

G5 CHECK THE TOW/HAUL INPUT CIRCUIT FOR A SHORT TO GROUND

- Disconnect: PCM [C175T](#).
- Disconnect: PCM [C1381T](#).
- Disconnect: PCM [C1551B](#).
- Inspect the connector for damaged, pushed-out terminals, corrosion or loose wires.
- Measure the resistance between the selector lever [C3245](#) Pin 7, circuit CET34 (BN/GN) and ground.



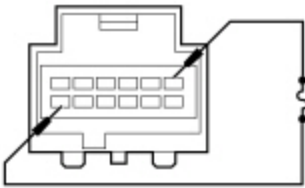
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Is the resistance greater than 10,000 ohms?

Yes	GO to G6 .
No	REPAIR circuit CET34 (BN/GN) for a short to ground. CLEAR the DTC. RERUN the KOER self-test and test the system for normal operation.

G6 CHECK THE TOW/HAUL INPUT CIRCUIT FOR AN OPEN

- Connect the scan tool.
- Enter the following diagnostic mode on the scan tool: TCS_DEPRES .
- Connect a fused jumper from the selector lever switch, [C3245](#) Pin 6 circuit CBP42 (GN) to [C3245](#) Pin 7 circuit CET34 (BN/GN) harness side.



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Does the TCS_DEPRES PID display YES?

Yes	REPLACE the selector lever. REFER to Selector Lever — Floor Shift in this section. CLEAR the DTC. RERUN the KOER self-test and test the system for normal operation.
No	INSPECT and REPAIR circuit CET34 (BN/GN) for an open. If an open circuit is not found, REPLACE the PCM. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Section 307-01. RERUN the KOER self-test and test the system for normal operation.

PINPOINT TEST H : THE SELECTSHIFT™ SWITCH IS NOT OPERATING CORRECTLY

NOTE: Vehicles equipped with a 3.7L engine use PCM [C175B](#).

NOTE: Vehicles equipped with a 5.0L or 6.2L engine use PCM [C1381B](#).

NOTE: Vehicles equipped with a 3.5L engine use PCM [C1551B](#).

H1 CHECK DTCS

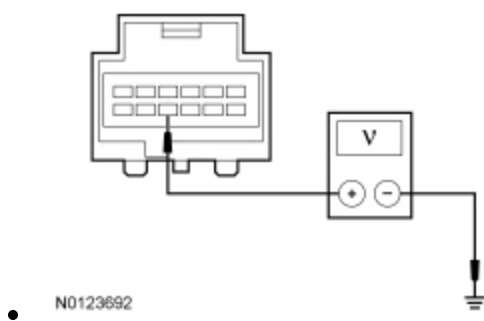
- Ignition ON.
- Connect the scan tool.
- Start the engine and RUN the Key ON Engine Running (KOER) self-test.

Is DTC P0815 set?

Yes	GO to H2 .
No	GO to H5 .

H2 CHECK THE UPSHIFT CIRCUIT FOR VOLTAGE

- Ignition OFF.
- Disconnect: Selector Lever [C3245](#).
- Inspect [C3245](#) for damaged, pushed-out terminals, corrosion or loose wires.
- Ignition ON.
- Measure the voltage between the selector lever [C3245](#) Pin 9, circuit CET43 (GY), harness side and ground.

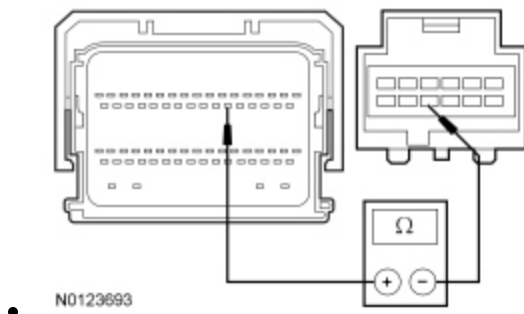


Is the voltage greater than 10 volts?

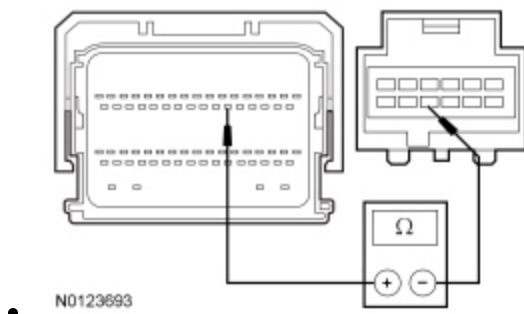
Yes	GO to H4 .
No	GO to H3 .

H3 CHECK THE UPSHIFT CIRCUIT FOR AN OPEN

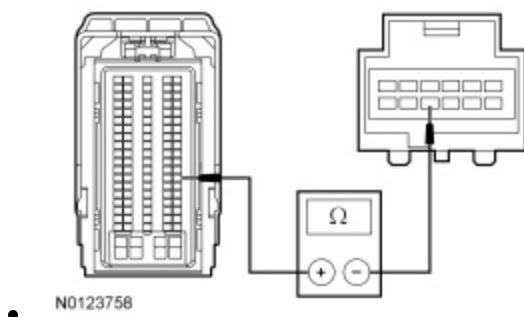
- Ignition OFF.
- Disconnect: PCM [C175B](#).
- Disconnect: PCM [C1381B](#).
- Disconnect: PCM [C1551B](#).
- Inspect the connector for damaged, pushed-out terminals, corrosion or loose wires.
- For 3.7L engine vehicles, measure the resistance between the Tow/Haul switch [C3245](#) Pin 9, circuit CET43 (GY) and PCM [C175B](#) Pin 23, harness side.



- For 5.0L or 6.2L engine vehicles, measure the resistance between the Tow/Haul switch [C3245](#) Pin 9, circuit CET43 (GY) and PCM [C1381B](#) Pin 23, harness side.



- For 3.5L engine vehicles, measure the resistance between the Tow/Haul switch [C3245](#) Pin 9, circuit CET43 (GY) and PCM [C1551B](#) Pin 89, harness side.

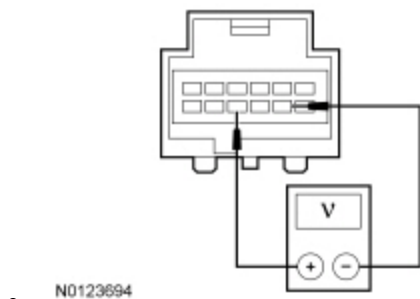


Is the resistance less than 5 ohms?

Yes	GO to H4 .
No	REPAIR circuit CET43 (GY) for an open. CLEAR the DTCs. RERUN the KOER self-test and test the system for normal operation.

H4 CHECK THE SIGNAL RETURN CIRCUIT FOR AN OPEN

- Connect: PCM [C175B](#).
- Connect: PCM [C1381B](#).
- Connect: PCM [C1551B](#).
- Ignition ON.
- Measure the voltage between the selector lever [C3245](#) Pin 9, circuit CET43 (GY) and [C3245](#) Pin 11, circuit RE407 (YE/VT) harness side.



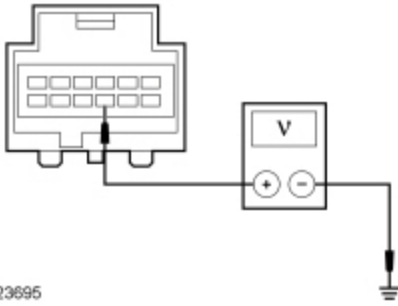
Is the voltage greater than 10 volts?

Yes	INSTALL a new selector lever, REFER to Selector Lever — Floor Shift in this section. CLEAR the DTCs. RERUN the KOER self-test and test the system for normal operation.
No	REPAIR the signal return circuit RE407 (YE/VT) for an open. If an open circuit is not found, REPLACE the PCM. PROGRAM the PCM with the latest calibration level. PERFORM the solenoid strategy download procedure, REFER to solenoid body strategy in Section 307-01. RERUN the KOER self-test and test the system for normal operation.

H5 CHECK THE DOWNSHIFT CIRCUIT FOR VOLTAGE

- Ignition OFF.
- Disconnect: Selector Lever [C3245](#).

- Inspect [C3245](#) for damaged, pushed-out terminals, corrosion or loose wires.
- Ignition ON.
- Measure the voltage between the selector lever [C3245](#) Pin 10, circuit CET42 (GN/VT), harness side and ground.

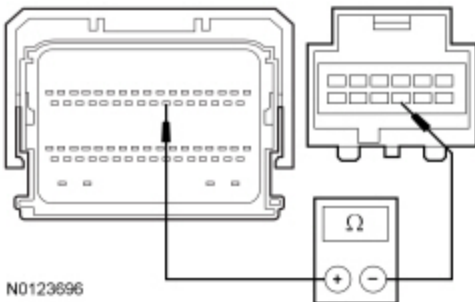


Is the voltage greater than 10 volts?

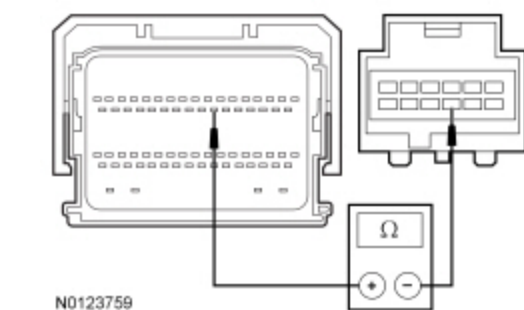
Yes	GO to H7 .
No	GO to H6 .

H6 CHECK THE DOWNSHIFT CIRCUIT FOR AN OPEN

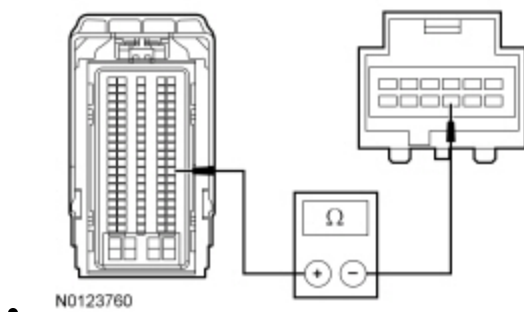
- Ignition OFF.
- Disconnect: PCM [C175B](#).
- Disconnect: PCM [C1381B](#).
- Disconnect: PCM [C1551B](#).
- Inspect the connector for damaged, pushed-out terminals, corrosion or loose wires.
- For 3.7L engine vehicles, measure the resistance between the selector lever [C3245](#) Pin 10, circuit CET42 (GN/VT) and PCM [C175B](#) Pin 24, harness side.



- For 5.0L or 6.2L engine vehicles, measure the resistance between the Tow/Haul switch [C3245](#) Pin 10, circuit CET42 (GN/VT) and PCM [C1381B](#) Pin 24, harness side.



- For 3.5L engine vehicles, measure the resistance between the Tow/Haul switch [C3245](#) Pin 10, circuit CET42 (GN/VT) and PCM [C1551B](#) Pin 88, harness side.

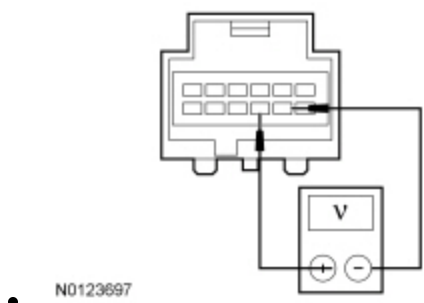


Is the resistance less than 5 ohms?

Yes	GO to H7 .
No	REPAIR circuit CET42 (GN/VT) for an open. CLEAR the DTCs. RERUN the KOER self-test and test the system for normal operation.

H7 CHECK THE SIGNAL RETURN CIRCUIT FOR AN OPEN

- Connect: PCM [C175B](#).
- Connect: PCM [C1381B](#).
- Connect: PCM [C1551B](#).
- Measure the voltage between the selector lever [C3245](#) Pin 10, circuit CET42 (GN/VT) and [C3245](#) Pin 11, circuit RE407 (YE/VT) harness side.



Is the voltage greater than 10 volts?

Yes	INSTALL a new selector lever, REFER to Selector Lever — Floor Shift in this section. CLEAR the DTCs. RERUN the KOER self-test and test the system for normal operation.
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No	INSPECT and REPAIR the signal return circuit RE407 (YE/VT) for an open. If an open circuit is not found, REPLACE the PCM. PROGRAM the PCM with the latest calibration level. PERFORM the solenoid strategy download procedure, REFER to solenoid body strategy in Section 307-01. RERUN the KOER self-test and test the system for normal operation.
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